

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1110

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture.
(BL3)

Assessment Tool Weightage: 40%

QUESTIONS: Any 4

Total Marks:40

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

*I DHAANUSH RAJAN - 192421162 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

ANALYTICAL PROBLEM SOLVING

1. Use Case Diagram Problem

A **Hospital Management System** allows patients to book appointments, doctors to update medical records, and administrators to manage hospital resources. The requirements are not clearly documented.

Tasks:

- a) Identify all actors involved in the system.
- b) List the primary use cases.
- c) Define relationships (include, extend, generalization).
- d) Construct a complete Use Case Diagram representing system functionality.

(10 Marks)

ANSWER

a) Actors Involved

1. **Patient** – Books appointments, views appointment details, checks medical records.
2. **Doctor** – Views appointments, updates patient medical records.
3. **Administrator** – Manages hospital resources, doctors, patients, and appointments.

b) Primary Use Cases

Patient

- Register/Login
- Book Appointment
- View Appointment
- Cancel Appointment
- View Medical Records

Doctor

- Login
- View Appointment Schedule
- Update Medical Records
- View Patient History

Administrator

- Login
- Manage Doctors
- Manage Patients

- Manage Appointments
- Manage Hospital Resources
- Generate Reports

c) Relationships

<<include>> Relationships

These are mandatory actions that are always performed.

- Book Appointment <<**include**>> Login
- Cancel Appointment <<**include**>> Login
- View Medical Records <<**include**>> Login
- Update Medical Records <<**include**>> View Patient History
- Manage Appointments <<**include**>> Login
- Manage Doctors <<**include**>> Login
- Manage Patients <<**include**>> Login
- Manage Hospital Resources <<**include**>> Login

<<extend>> Relationships

These are optional or conditional actions.

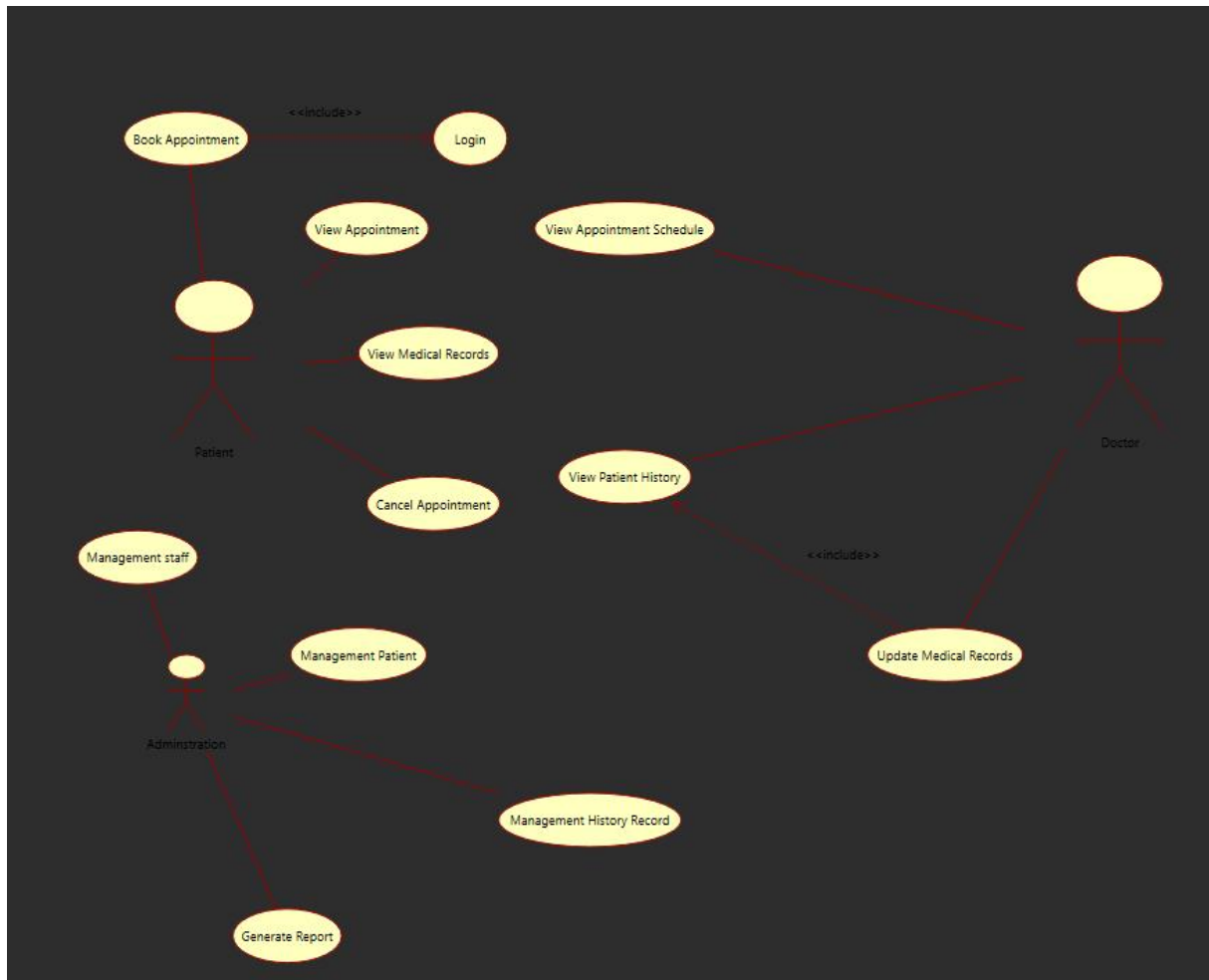
- Cancel Appointment <<**extend**>> Book Appointment
- Generate Reports <<**extend**>> Manage Hospital Resources

Generalization

- **User** (*General Actor*)
 - Patient
 - Doctor
 - Administrator

All three actors inherit common functionality such as **Login**.

d)



2. Class Diagram Problem

A **Library Management System** contains entities such as Book, Member, Librarian, and IssueRecord. The relationships among entities are not properly defined.

Tasks:

- a) Identify classes with attributes and methods.
- b) Define relationships (association, aggregation, inheritance).
- c) Specify multiplicity between classes.
- d) Design a detailed Class Diagram for the system.

(10 Marks)

ANSWER

a) Classes with Attributes and Methods

1. Book

Attributes:

- bookId : int
- title : String
- author : String
- publisher : String
- isbn : String
- status : String (Available/Issued)

Methods:

- addBook()
- updateBook()
- searchBook()
- checkAvailability()

2. Member

Attributes:

- memberId : int
- name : String
- address : String
- phone : String
- email : String

Methods:

- borrowBook()
- returnBook()

- viewIssuedBooks()
-

3. Librarian

Attributes:

- librarianId : int
- name : String
- employeeId : String

Methods:

- issueBook()
- receiveBook()
- addBook()
- removeBook()

4. IssueRecord

Attributes:

- issueId : int
- issueDate : Date
- dueDate : Date
- returnDate : Date
- fine : double

Methods:

- calculateFine()
- updateReturnDate()
- generateRecord()

b) Relationships

1. Association

- Member ↔ IssueRecord
- Book ↔ IssueRecord
- Librarian ↔ IssueRecord

1. Aggregation

- Librarian manages many Books.
- Library (optional class) aggregates Books, Members, and Librarians.

1. Inheritance (Optional)

- Person (Superclass)
 - Member (Subclass)
 - Librarian (Subclass)

c) Multiplicity

Relationship

Multiplicity

Member → IssueRecord 1 Member can have **0..*** IssueRecords

Book → IssueRecord 1 Book can appear in **0..*** IssueRecords (over time)

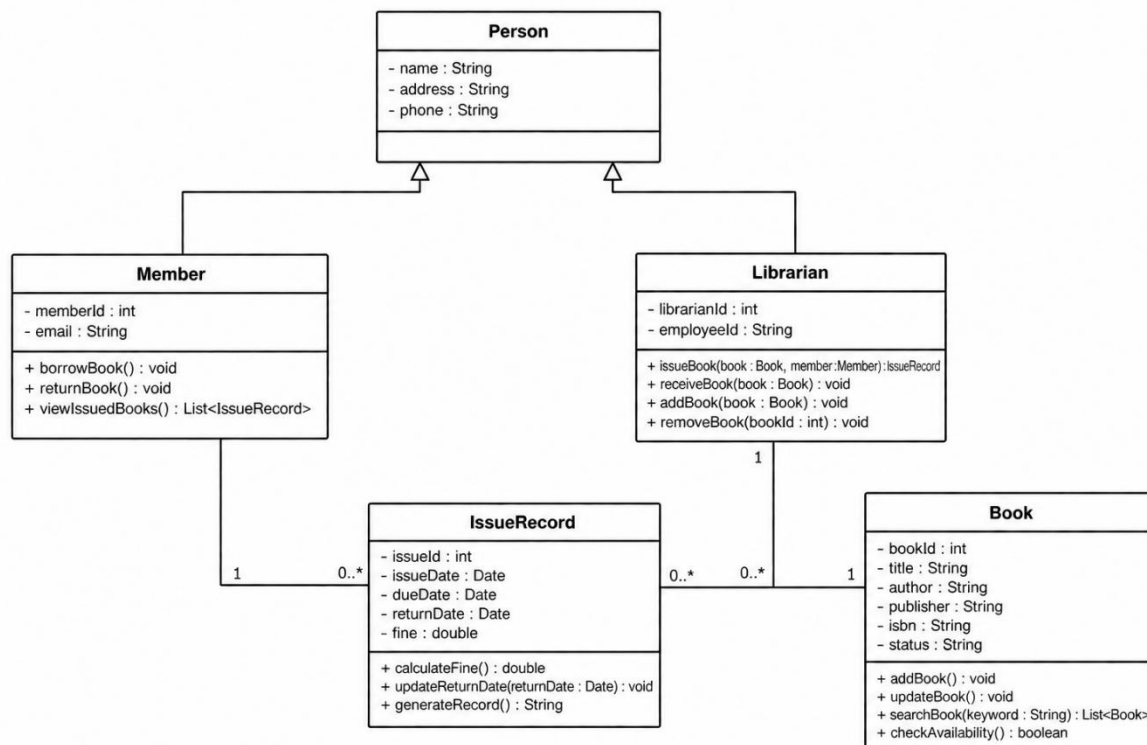
Librarian → IssueRecord 1 Librarian manages **0..*** IssueRecords

IssueRecord → Member Each IssueRecord belongs to **1** Member

IssueRecord → Book Each IssueRecord refers to **1** Book

IssueRecord → Librarian Each IssueRecord is handled by **1** Librarian

d)



3. Sequence Diagram Problem

In an **Online Food Ordering System**, a customer selects food items, places an order, makes payment, and receives order confirmation. The interaction flow is unclear.

Tasks:

- a) Identify objects involved in the process.
- b) Define message flow between objects.
- c) Represent activation and lifelines.
- d) Construct a Sequence Diagram showing complete interaction flow.

(10 Marks)

ANSWER

(a) Objects Involved

The following objects participate in the sequence:

- **Customer**
- **Food Ordering System**
- **Menu**
- **Order**
- **Payment Gateway**
- **Confirmation Service**

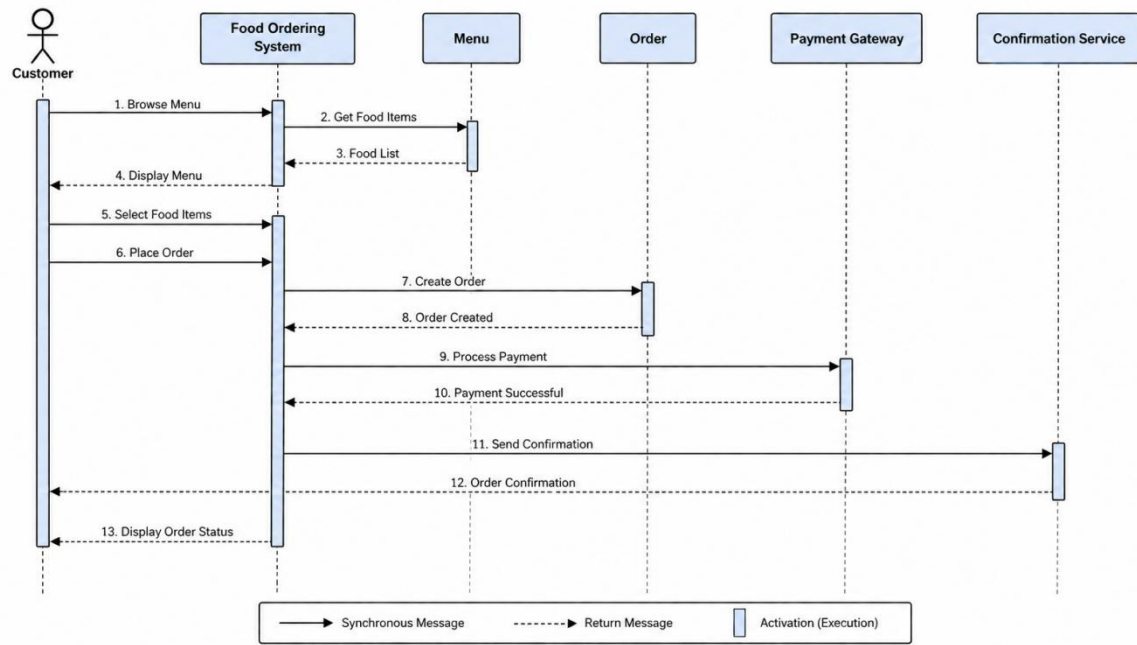
(b) Message Flow Between Objects

1. Customer → Food Ordering System: Login / Browse Menu
2. Food Ordering System → Menu: Display Food Items
3. Customer → Food Ordering System: Select Food Items
4. Customer → Food Ordering System: Place Order
5. Food Ordering System → Order: Create Order
6. Food Ordering System → Payment Gateway: Process Payment
7. Payment Gateway → Food Ordering System: Payment Successful
8. Food Ordering System → Confirmation Service: Send Confirmation
9. Confirmation Service → Customer: Order Confirmation
10. Food Ordering System → Customer: Display Order Status

(c) Activation and Lifelines

- **Lifelines:** Customer, Food Ordering System, Menu, Order, Payment Gateway, Confirmation Service.
- **Activation Bars:** Represent the period during which each object is performing an operation.
- **Messages:** Represent communication between the objects in chronological order from top to bottom.

d)



4. Activity Diagram Problem

A **Student Admission System** includes application submission, document verification, fee payment, and enrollment confirmation. The workflow lacks proper decision-making steps.

Tasks:

- a) Identify all activities in the workflow.
- b) Add decision nodes for approval/rejection.
- c) Include parallel processing where applicable.
- d) Design an optimized Activity Diagram.

(10 Marks)

ANSWER

- (a) Identify all activities in the workflow

The activities involved in the Student Admission System are:

1. Start
2. Submit Application
3. Upload Required Documents
4. Verify Documents
5. Check Eligibility
6. Pay Admission Fee
7. Verify Payment
8. Process Enrollment
9. Generate Enrollment ID
10. Send Admission Confirmation
11. Update Student Record
12. Admission Confirmed
13. End

- (b) Add decision nodes for approval/rejection

The following decision nodes are included:

- **Documents Valid?**
 - **Yes** → Proceed to Eligibility Check
 - **No** → Request Additional Documents → Re-submit Documents
- **Eligible?**
 - **Yes** → Continue with Admission Process
 - **No** → Reject Application → Notify Student → End
- **Payment Successful?**
 - **Yes** → Continue to Enrollment
 - **No** → Retry Payment

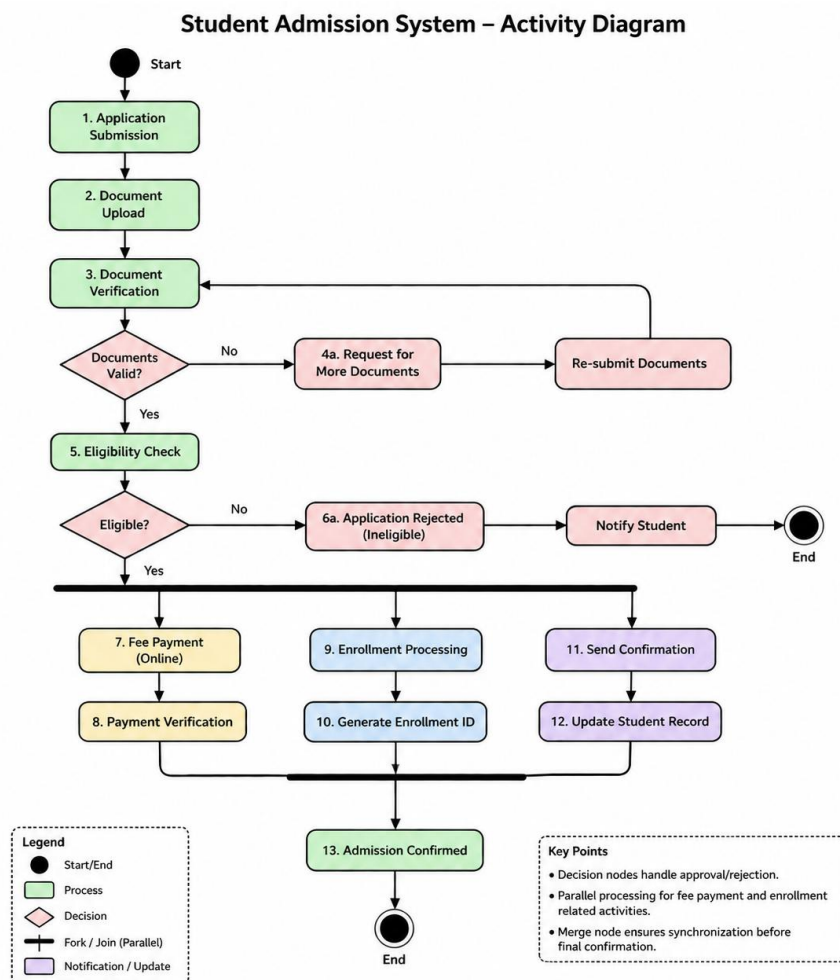
(c) Include parallel processing where applicable

After the student is found eligible, the following activities can occur **in parallel (Fork Node)**:

- Fee Payment and Payment Verification
- Enrollment Processing and Enrollment ID Generation
- Student Record Update

After all parallel activities are completed, they are synchronized using a **Join Node**, followed by Admission Confirmation.

d)



RUBRICS FOR CALCULATION

ANALYTICAL PROBLEM SOLVING

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation